Nonparametric Filters

Probabilistic Robotics, chapter 4

Lesson t2.3 · Thrun, Burgard, Fox

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Why drop the Gaussian?

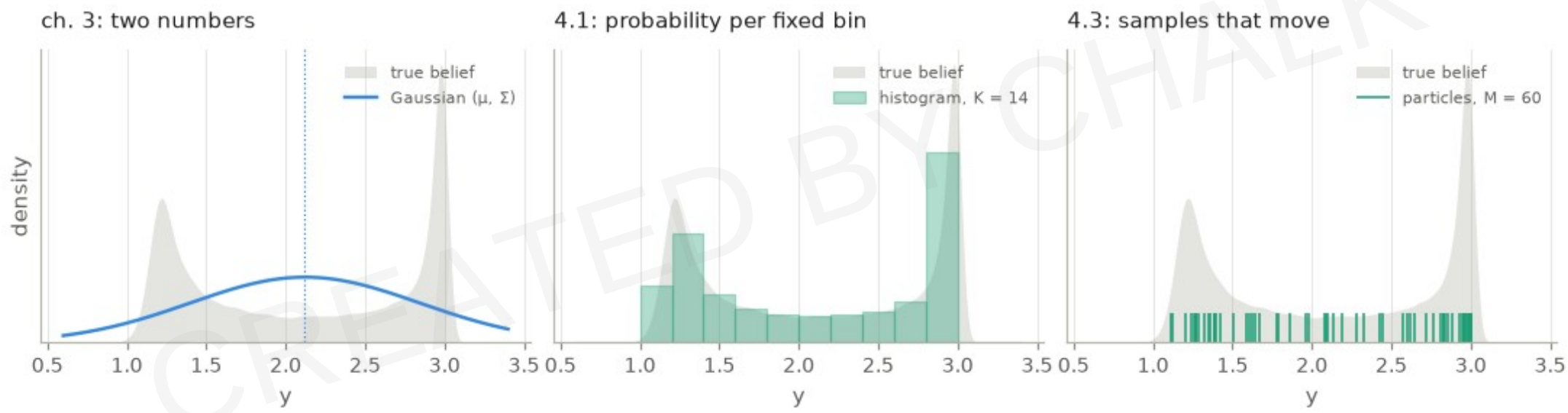
- A Gaussian forced onto a two-humped belief puts its mean between the humps
- In the figure that mean sits where the true density is 0.097 of its peak
- A field with point symmetry gives a robot two equally good poses
- A nonparametric filter stores a finite set of numbers: bins or samples
- The price is computation, which grows with the state dimension

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One belief, three representations

Fig L1 · ch. 4

The Gaussian puts its mean where the belief has almost no mass



Gaussian (ch. 3), histogram (4.1), particles (4.3)

Two numbers cannot hold two humps; bins and samples can

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Roadmap

1. 4.1 Histogram filter: fixed bins, one probability per bin
2. 4.2 Binary Bayes filter: one static bit, in log odds
3. 4.3 Particle filter: moving samples, weights, resampling
4. 4.3.2 and 4.3.3 Why weighting by the likelihood is correct
5. 4.3.4 How a finite set lies: variance, diversity, bias, deprivation

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A robot on a ring corridor

Fig L2 · Table 4.1

Doors seen 2.5 m apart: only the 4 m hypothesis explains both; line 3 is a shift plus a blur



- Predict: shift plus blur
- Update: multiply, then normalize

Doors at 1.5, 4.0 and 7.5 m; sees a door, drives 2.5 m, sees a door

Only the hump that explains both sightings survives

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Discrete Bayes filter, Table 4.1

$$\bar{p}_{k,t} = \sum_i p(X_t = x_k \mid u_t, X_{t-1} = x_i)\, p_{i,t-1} \qquad \text{line 3}$$

$$p_{k,t} = \eta\, p(z_t \mid X_t = x_k)\, \bar{p}_{k,t} \qquad \text{line 4}$$

$\bar{p}_{k,t}$ predicted mass in bin k

$p_{i,t-1}$ mass in bin i one step ago

η normalizer: one over the sum

x_k the state of bin k

- Line 3 over all k is a matrix-vector product $\bar{p} = T\, p$, columns of T sum to 1
- Line 4 is an element-wise product, then division by the sum
- In signal processing this is the forward pass of a hidden Markov model

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Predict: where is the mass after the second door?

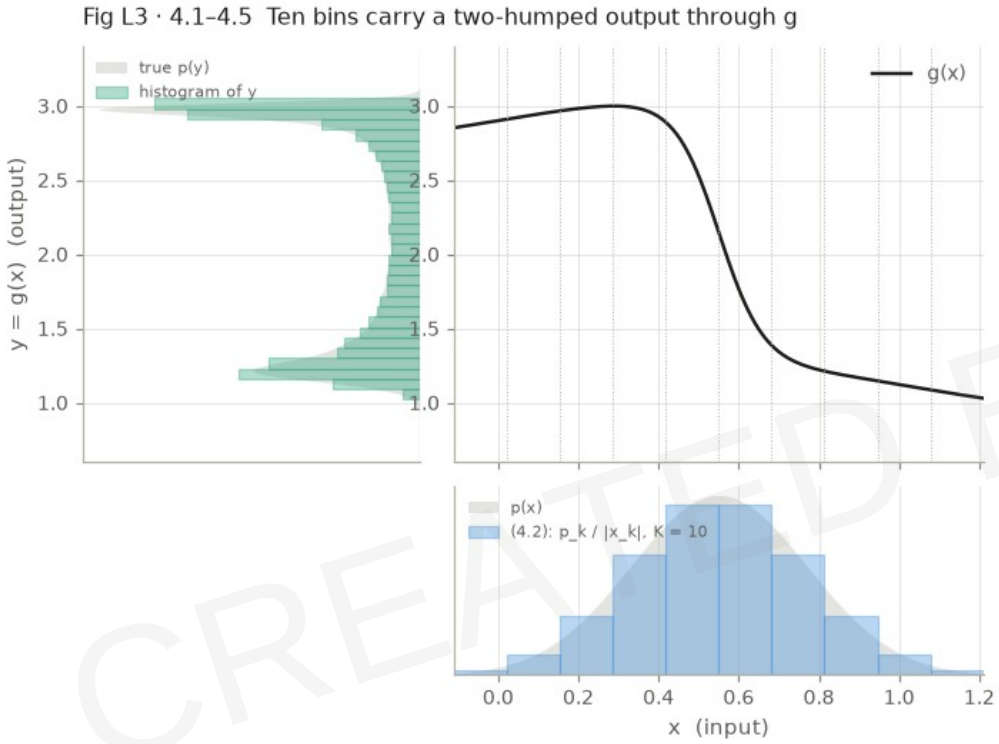
- Ring corridor, 100 bins of 0.1 m, doors at 1.5, 4.0 and 7.5 m
- Uniform prior, door seen, drive 2.5 m, door seen again

?

What fraction of the mass lies within 1 m of 4.0 m, and why is it not close to 1?

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Histograms over a continuous state



- Error is the bin width, not the nonlinearity
- Total variation: 0.066 at 5 bins, 0.021 at 10, 0.006 at 40

A Gaussian cut into bins, pushed through a nonlinear g

Halve the bin width before you blame the nonlinearity

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Piecewise constant density and the bin approximations

$$p(x_t) = \frac{p_{k,t}}{|\mathbf{x}_{k,t}|} \quad \text{for } x_t \in \mathbf{x}_{k,t} \tag{4.2}$$

$$p(z_t \mid \mathbf{x}_{k,t}) \approx p(z_t \mid \hat{x}_{k,t}) \tag{4.4}$$

$$p(\mathbf{x}_{k,t} \mid u_t, \mathbf{x}_{i,t-1}) \approx \eta\, |\mathbf{x}_{k,t}|\, p(\hat{x}_{k,t} \mid u_t, \hat{x}_{i,t-1}) \tag{4.5}$$

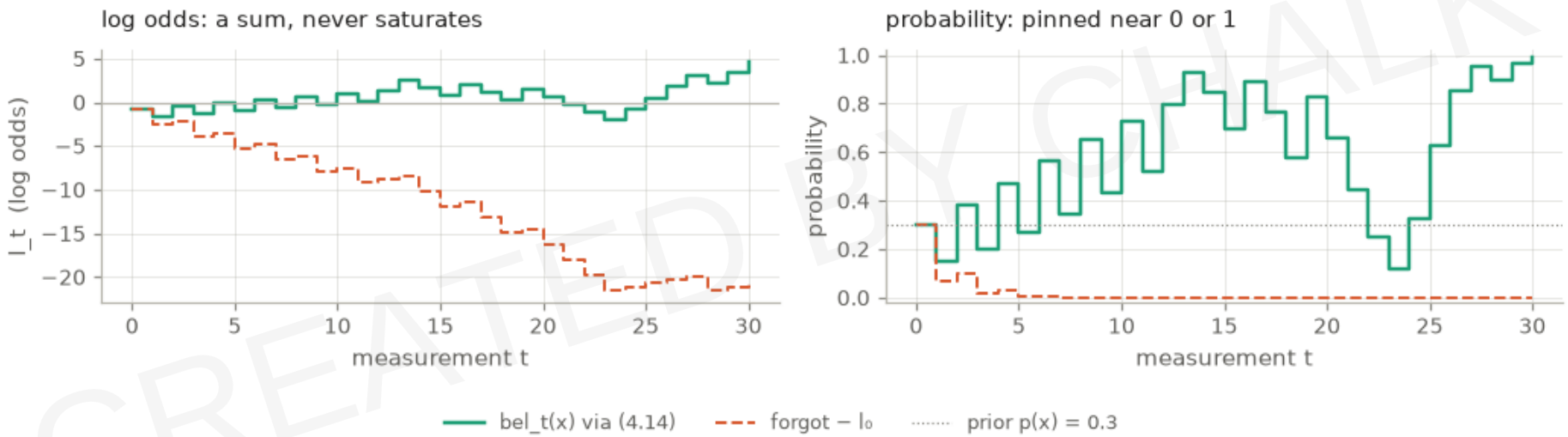
- $\hat{x}$ is the mean of the bin (4.3); probe every model at that point
- The η of (4.5) normalizes each column of T

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Binary Bayes filter: log odds as a running sum

Fig L4 · Table 4.2

Every update adds the inverse model's log odds and takes the prior's back out



- Fifteen hits in thirty give bel = 0.990
- Forgetting the prior gives bel = 0.000

Prior 0.3, thirty readings; dashed: prior term forgotten

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Table 4.2: the log-odds update

$$l_t = l_{t-1} + \log \frac{p(x \mid z_t)}{1 - p(x \mid z_t)} - \log \frac{p(x)}{1 - p(x)} \qquad \text{line 2}$$

$$bel_t(x) = 1 - \frac{1}{1 + \exp\{l_t\}} \qquad (4.14)$$

l_t log odds of the cell after t readings

p(x $_t$ z□) inverse measurement model

p(x) prior; its log odds is l_0

- $p(x \mid z_t)$ is an inverse measurement model: state given reading
- Subtract l_0 every update; code it with expit for stability

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Predict: forget the prior, then what?

- A cell with prior 0.3, fifteen hits in thirty readings; correct code ends at bel = 0.990
- Drop the subtracted prior term from line 2

? **Where does bel end up, and why?**

A Still near 0.990

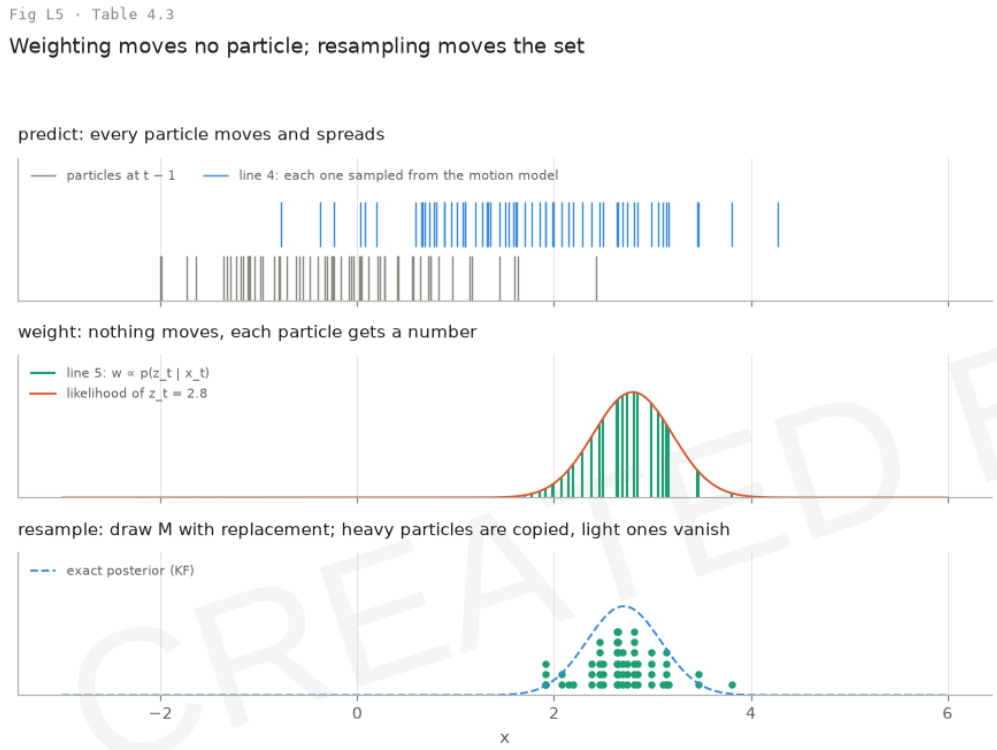
B Pinned near 0

C Pinned near 1

D Stuck at the prior 0.3

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One pass of the particle filter



- Mean 2.678 against the KF's 2.709
- Std 0.378 against 0.377

1-D linear-Gaussian system, so the Kalman filter gives the exact answer

On a linear-Gaussian system the particles agree with the Kalman filter

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Table 4.3: sample, weight, resample

1 Sample (line 4)

Draw $x_t^{[m]}$ from $p(x_t \mid u_t, x_{t-1}^{[m]})$, one draw per particle: the motion model of ch. 5

2 Weight (line 5)

Importance factor $w_t^{[m]} = p(z_t \mid x_t^{[m]})$: the sensor model of ch. 6

3 Resample (8–11)

Draw i with probability proportional to $w_t^{[i]}$, M times, with replacement

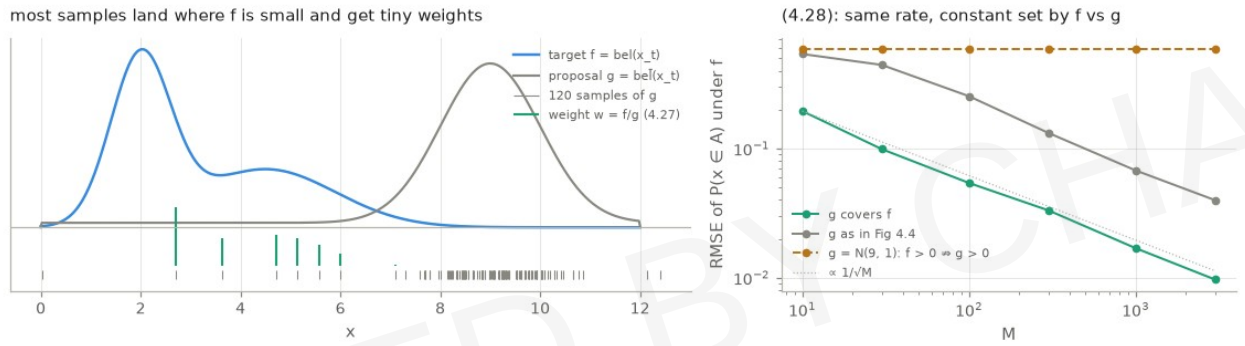
- The ideal posterior (4.23) holds only as $M \to \infty$

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Importance sampling: how well g covers f

Fig L6 · 4.25–4.28, Fig 4.4

Importance sampling is exact on average; the proposal decides how many samples it costs



- Covering proposal: RMSE 0.017 at M = 1000
- Book's proposal: 0.068, about sixteen times the samples
- A proposal that misses f returns 0 at every M

Target f, proposal g, weights f/g; error against M for three proposals

The cost of a proposal is measured in samples

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Importance sampling and its weights

$$E_f[I(x \in A)] = \int \frac{f(x)}{g(x)} g(x)\, I(x \in A)\, dx = E_g[w(x)\, I(x \in A)] \tag{4.25}$$

$$w^{[m]} = \frac{f(x^{[m]})}{g(x^{[m]})} \tag{4.27}$$

- Valid only if f(x) > 0 implies g(x) > 0
- In the PF, f = bel(x_t) and g is the motion model applied to bel(x_{t-1})

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Predict: how many samples does a bad proposal cost?

- Target f, estimate P(x in (0, 3)) = 0.587; both errors fall as 1/√M
- Proposal A covers f; proposal B is the book's, piled up at x = 9

? **By what factor must M grow for B to match A?**

A About 2

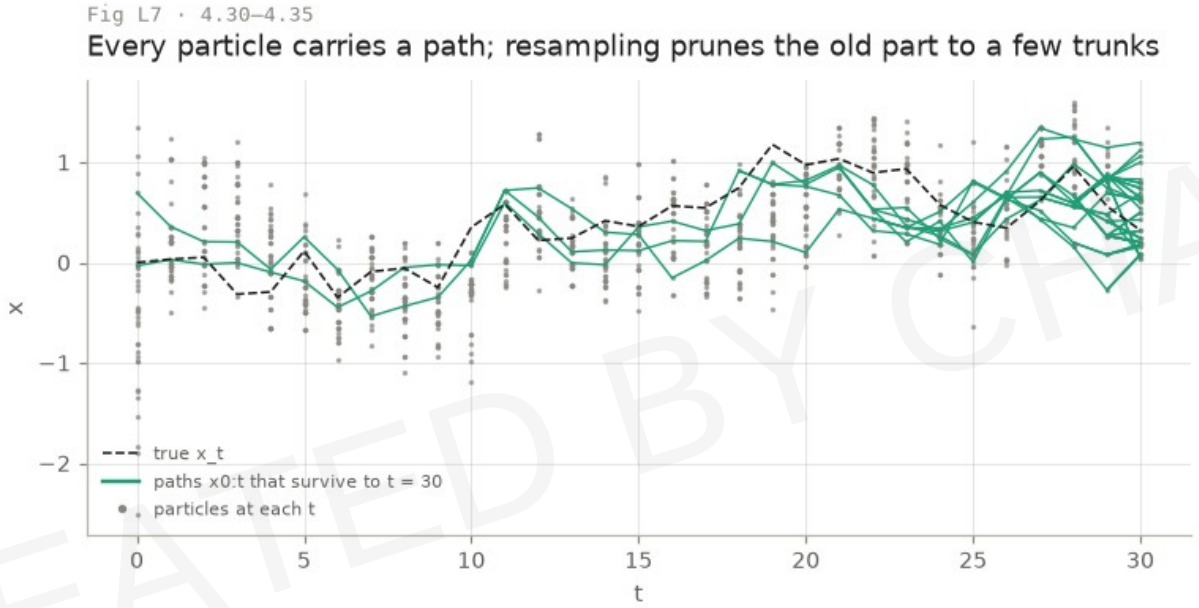
B About 4

C About 16

D B can never match A

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Particles as paths: the past collapses



- Of 40 particles at t = 0, only 2 have descendants at t = 30

Gray: particles per step; teal: histories of particles alive at t = 30

Resampling keeps the present diverse and the past thin

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Why the weight is the measurement likelihood

$$p(x_{0:t} \mid z_{1:t}, u_{1:t}) = \eta\, p(z_t \mid x_t)\, p(x_t \mid x_{t-1}, u_t)\, p(x_{0:t-1} \mid z_{1:t-1}, u_{1:t-1}) \tag{4.32}$$

$$w_t^{[m]} = \frac{\text{target}}{\text{proposal}} = \eta\, p(z_t \mid x_t) \tag{4.34}$$

- Motion term and previous-path term cancel between target and proposal
- Argument is correct only as M → ∞ (p. 104)

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The one idea of 4.3

A particle filter is importance sampling, repeated every time step

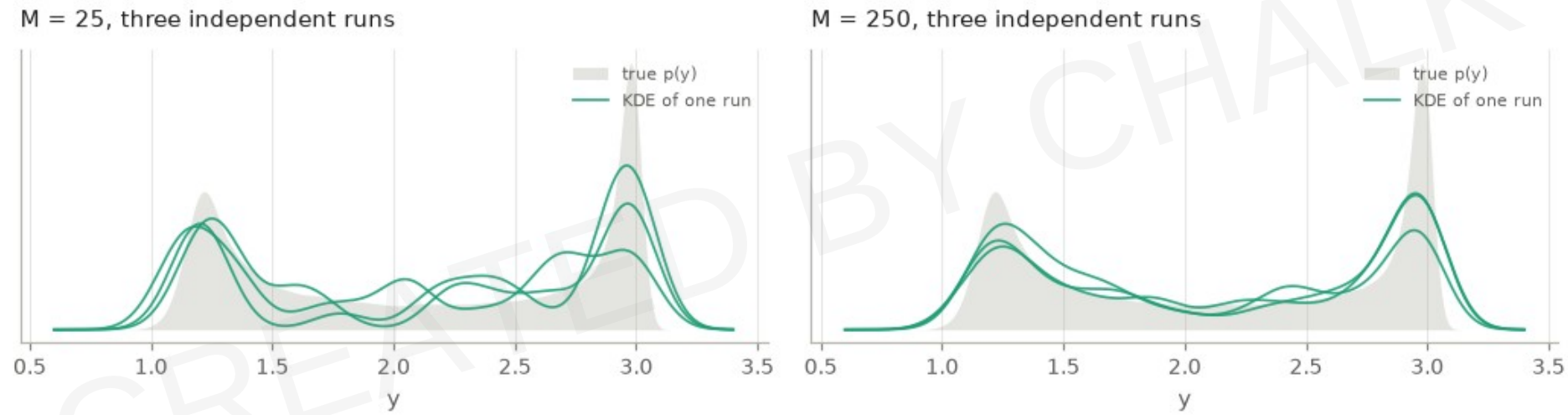
Proposal: the motion model applied to the old particles. Weight: the measurement likelihood.

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Sampling variance: the estimate is random

Fig L8 · Fig 4.6

Same belief, same g, different random numbers: the estimate is itself a random variable



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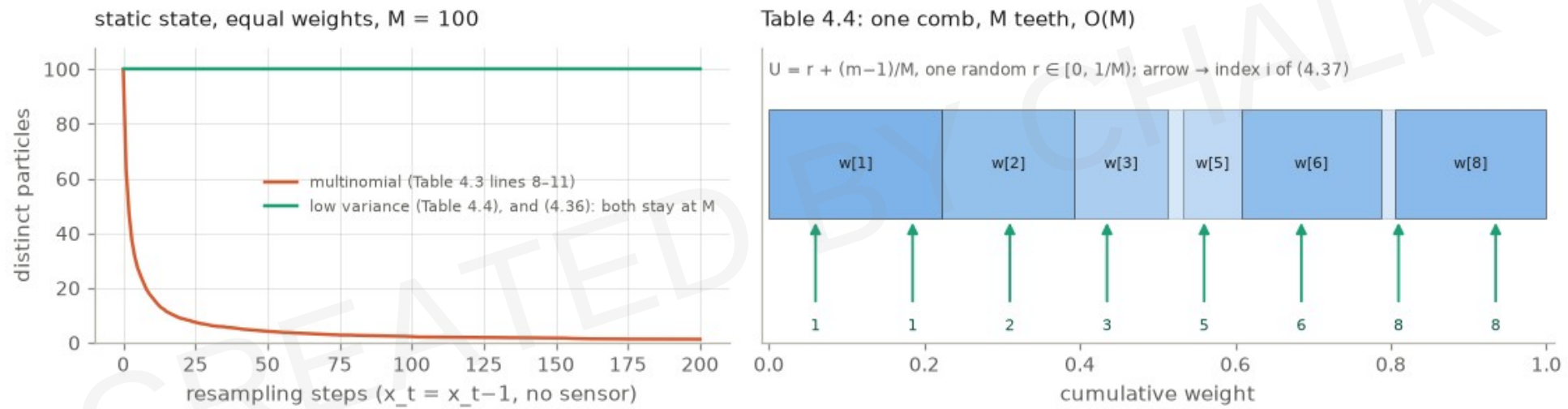
- Std of the sample mean: 0.146 at M = 25, 0.046 at M = 250
- Ratio 3.19, close to the square root of ten

Same belief, same g, 25 and 250 samples, three runs each

Resampling loses diversity

Fig L9 · (4.36), Table 4.4, Fig 4.7

Resampling with no information destroys diversity; the comb does it least



- M = 100, multinomial: 16.7 distinct after 10 steps, 2.3 after 100
- Low-variance sampler keeps all 100

Left: no motion, no sensor, resample every step; right: low-variance comb

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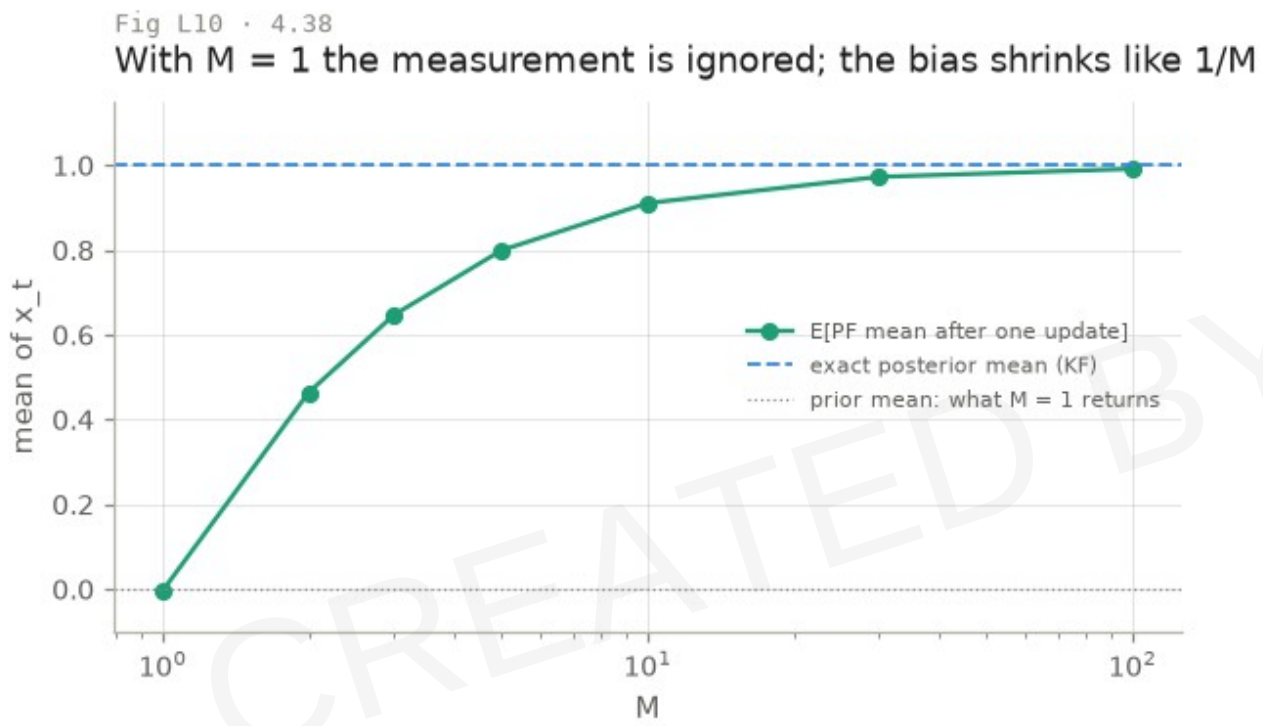
Resample less often; pick by cumulative weight

$$i = \operatorname{argmin}_j \sum_{m=1}^{j} w_t^{[m]} \geq U \tag{4.37}$$

- (4.36): reset weights to 1 after resampling, multiply them otherwise
- Decide by the variance of the weights; never resample a static state
- Table 4.4 assumes normalized weights; Table 4.3 never normalizes
- O(M) instead of O(M log M) for M binary searches

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Sampling bias at small M



- M = 1: −0.003, the measurement is ignored
- M = 2: 0.464; M = 10: 0.911; M = 100: 0.992

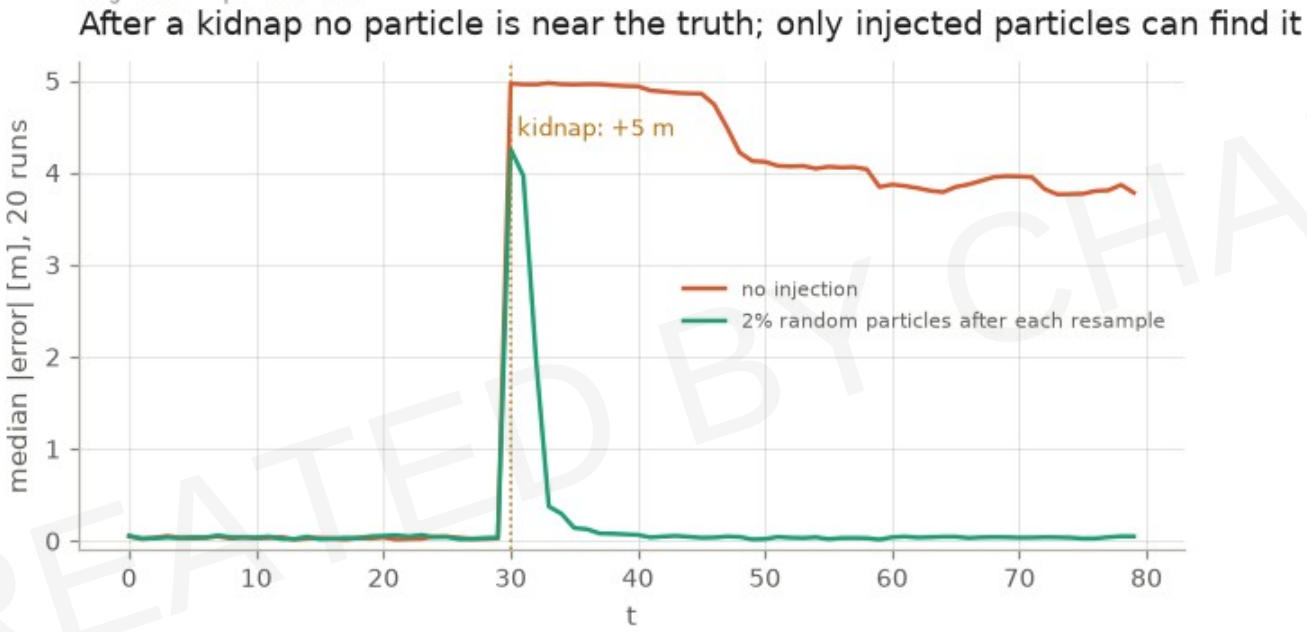
Expected PF mean over 40,000 runs; exact posterior mean is 1.0

More runs do not remove the bias; only more particles do

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Particle deprivation and random injection

Fig L11 · p. 112–113



- No injection: 0 of 20 runs recover
- 2% uniform injection: 20 of 20 recover, median 4 steps

Ring corridor, M = 300, robot moved 5 m at t = 30

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What a finite particle set costs

0.146

std of the mean at M = 25 (0.046 at 250)

2.3

distinct particles of 100 after 100 blind resamples

0.464

expected mean at M = 2; the exact answer is 1.0

0 of 20

runs recover from a kidnap without injection

- Variance, lost diversity, bias and deprivation: the four ways of 4.3.4

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Histogram or particles?

Histogram filter (4.1)

- Fixed bins over the whole state space
- Cost grows with the number of bins
- Error set by the bin width
- Exact sums, no sampling noise

Particle filter (4.3)

- Samples gather where the mass is
- Cost grows with M
- Variance, bias, lost diversity
- Needs injection to recover from a kidnap

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Both converge to the true posterior as the number of values grows

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Traps

Symptom	Cause	Fix
Mean between two plausible poses	Multimodal belief	Histogram or particles
Map drifts with no evidence	Prior not subtracted	Subtract l_0 every update
All weights zero or NaN	Likelihood underflow	Log weights, logsumexp
N_eff $\ll$ M after each update	Proposal wider than likelihood	Better proposal, more particles
Stationary robot falsely certain	Resampling without information	Suspend resampling
Sampler index out of range	Unnormalized weights	Normalize first
Confident and wrong after kidnap	Particle deprivation	Injection, augmented MCL

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Summary

- Table 4.1 is Table 2.1 with a sum; (4.5) needs a per-column η
- Table 4.2 adds evidence and subtracts the prior in log odds
- Table 4.3 weights by $p(z_t \mid x_t)$ because the motion term cancels (4.34)
- Importance sampling needs $f > 0 \Rightarrow g > 0$ and errors fall as $1/\sqrt{M}$
- Resample on weight variance with the low-variance sampler
- A finite set can lose diversity, carry bias at small M, or lose the truth

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What comes next

1. Ch. 5 and 6: motion and measurement models to sample from and weight with
2. Ch. 8: grid and Monte Carlo localization, augmented MCL
3. Ch. 9: occupancy grids, Table 4.2 run once per cell
4. Ch. 13: FastSLAM, particles that carry a map

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